AGU 2021-- U51B: Open Science in Action

Introducing: Kerchunk

Cloud-performant reading of NetCDF4/HDF5/Grib2 using the Zarr library

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SIParCS – Summer Internships in Parallel Computational Science – 2017 – BOULDER · CO

The Problem with Geophysical Data in the Cloud

- Many data archives being moved to the cloud
- Great for storage-adjacent computing
- Many datasets still in native NetCDF4/HDF5/GRIB format
- Other formats like Zarr are more cloud-optimized but require data conversion/duplication

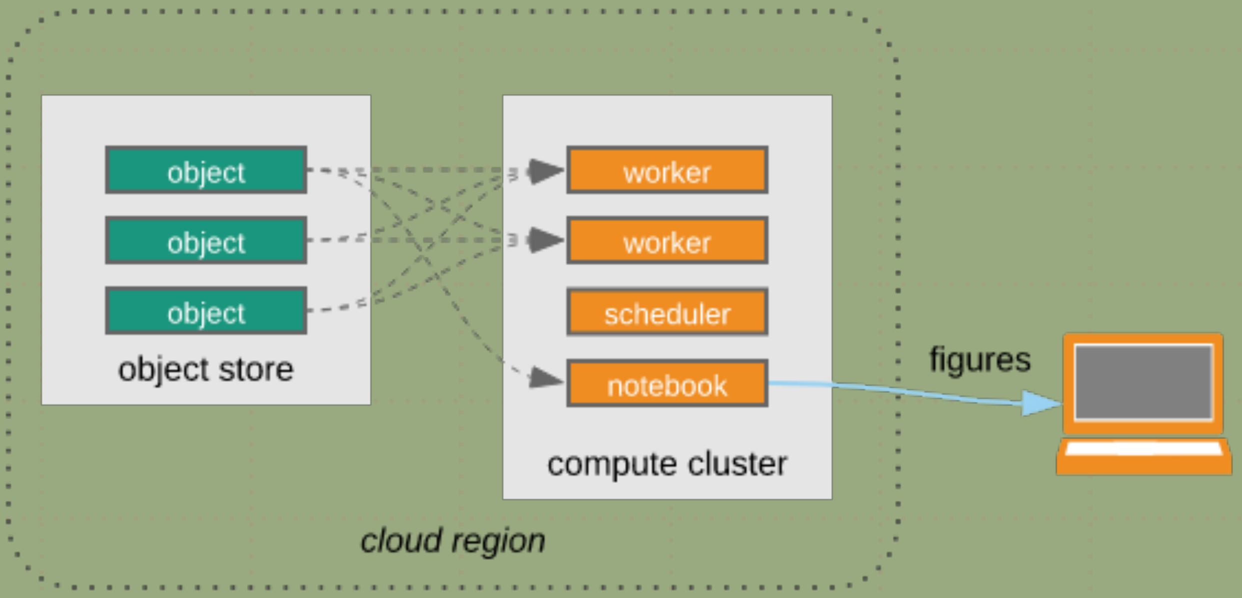


Image courtesy of Ryan Abernathey

HDF/NetCDF Metadata

- Reading metadata requires many small reads of the file
- Inefficient for accessing cloud-hosted data

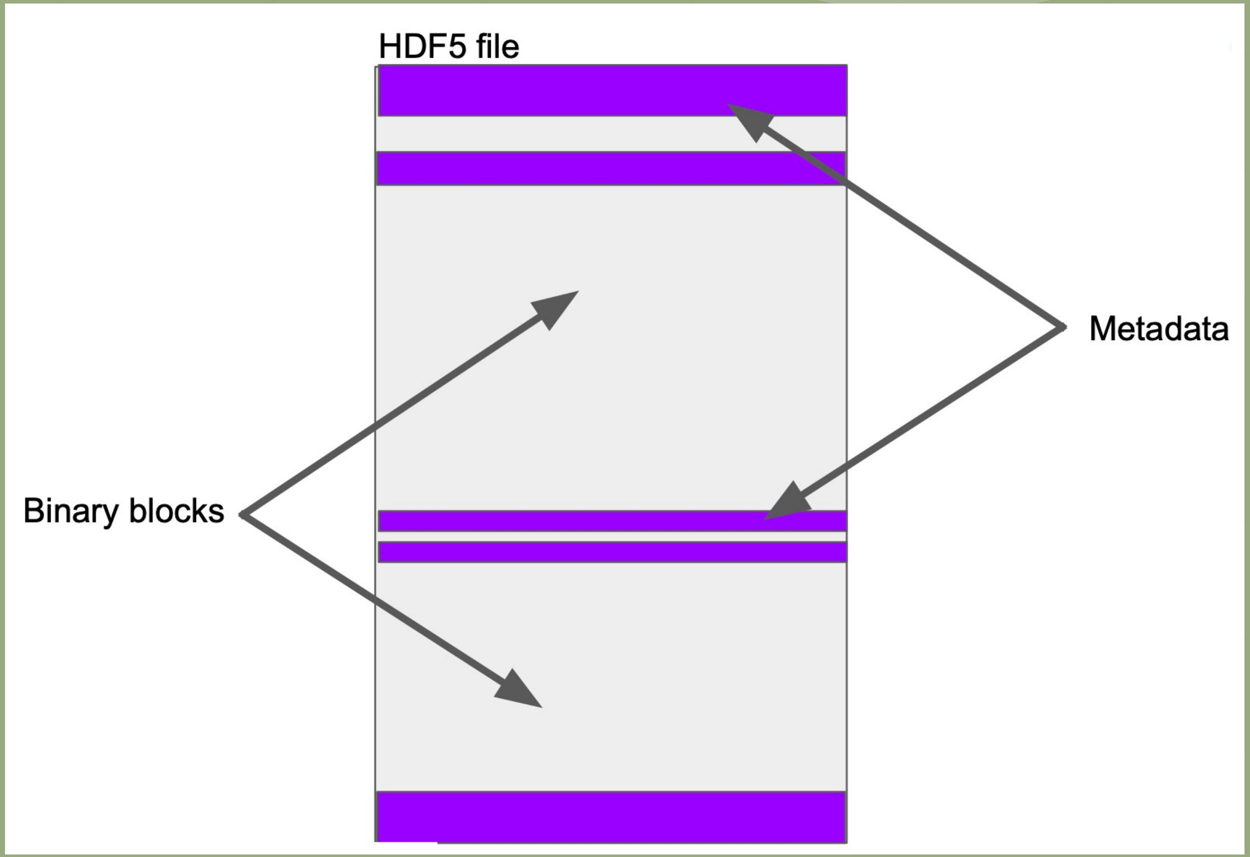
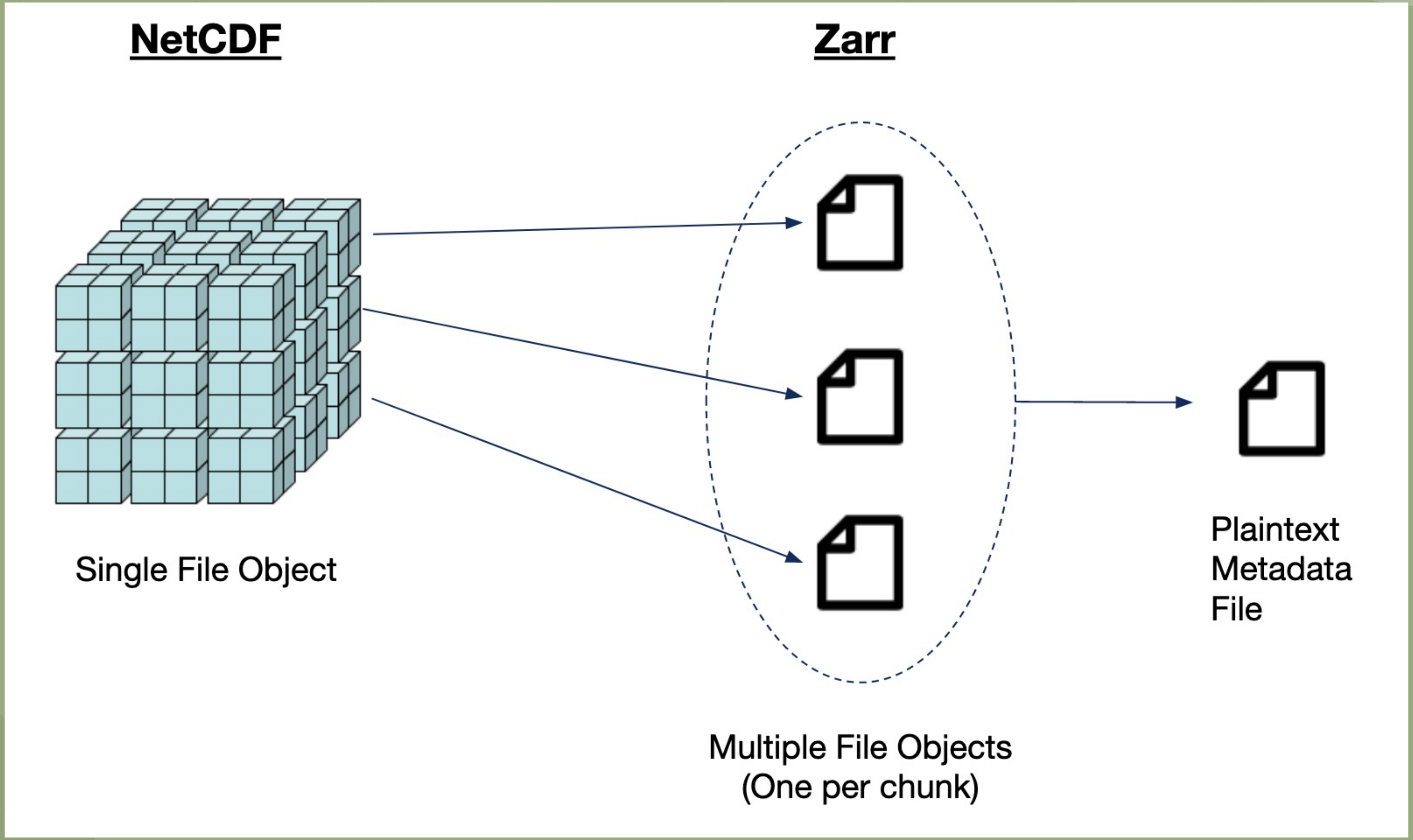


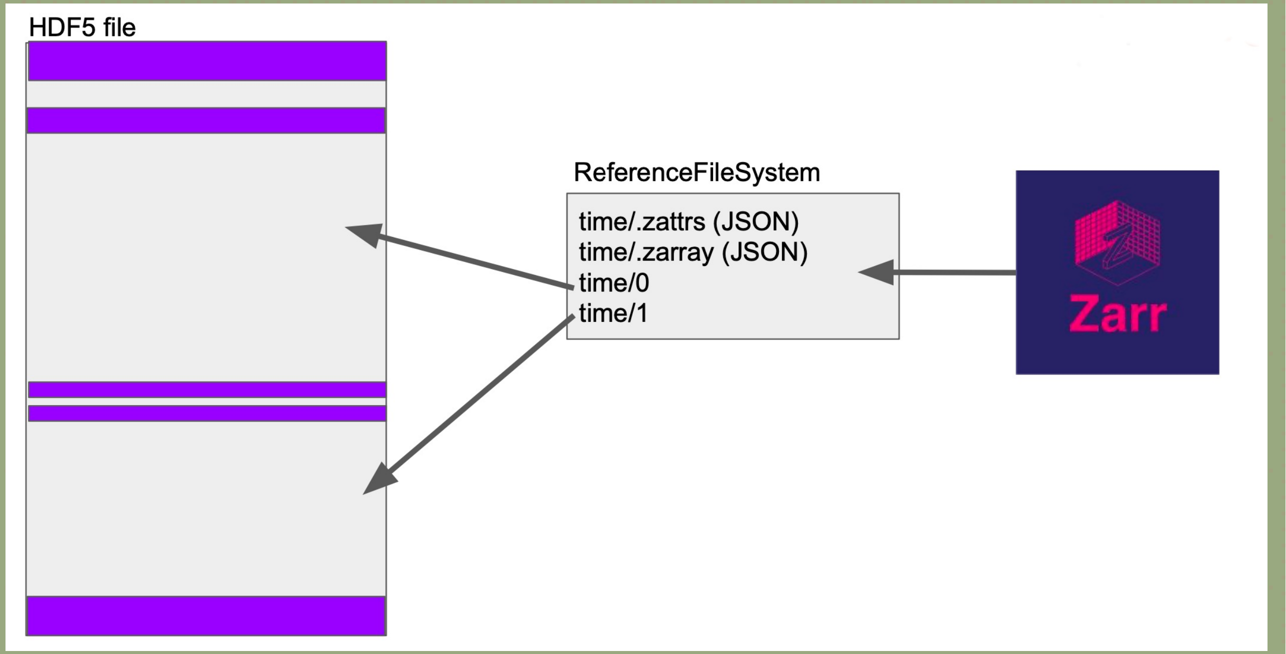
Image courtesy of Martin Durant

Zarr


NetCDF
Zarr
Single File Object
Plaintext
Metadata
File
Multiple File Objects
(One per chunk)

Introducing: Kerchunk

- Consolidates HDF/NetCDF/GRIB metadata into Zarr-spec JSON
- Metadata can be read without opening data file
- Data can be loaded in parallel
- Can Bridge multiple datasets



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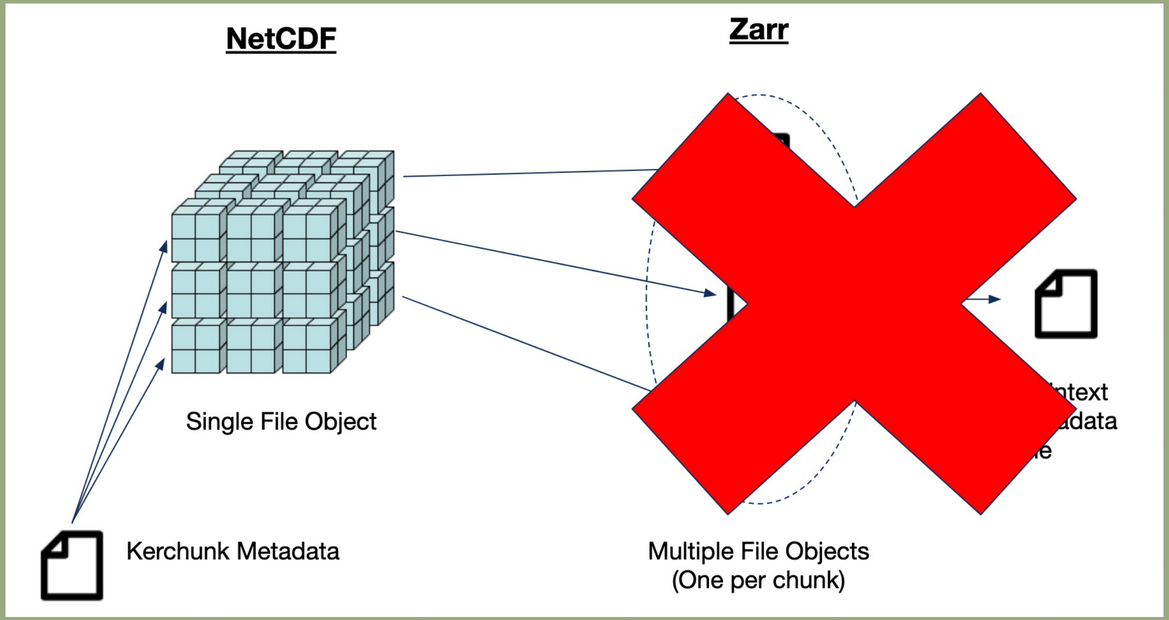

Introducing: Kerchunk

- Nearly as fast as Zarr itself
- Avoids need for data duplication
- Can be extended to work on many different data formats
    - Currently supports NetCDF, HDF5, GRIB
- Metadata files can be easily shared/hosted

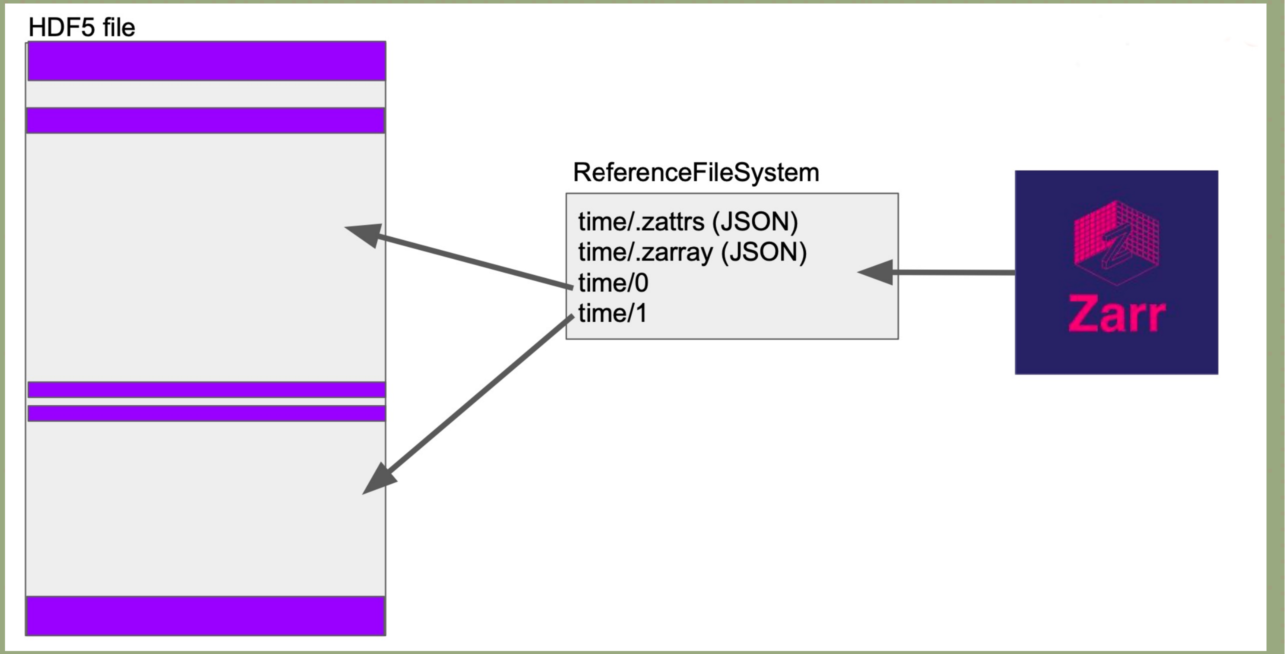

Image courtesy of Martin Durant

More Info

- Contact me
  - Email: lsterzinger@ucdavis.edu
  - Twitter @lucassterzinger

- Links to Slides, AGU Poster, Kerchunk, and more!
  - https://lucassterzinger.com/2021-agu-poster